

KIRAAT: A 3,105-Hour Turkish Read-Speech Corpus Segmented at Sentence Boundaries and Released Without Quality Gates

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Abstract

Turkish has abundant text and speech-recognition data but very little read speech suitable for training text-to-speech (TTS): the openly available read-speech corpora for the language are measured in tens or low hundreds of hours. We release KIRAAT, a corpus of 1,840,404 clips totalling 3,105.7 hours of Turkish read speech, segmented from 2,680 audiobook and spoken-article recordings across 27 channels, together with the pipeline that produced it.

Two design rules distinguish the pipeline from the established recipe. First, clips are cut at sentence boundaries recovered from word-level forced alignment, not at silences found by voice activity detection. In a controlled ablation over 206 recordings (240.5 hours) in which only the cut point varies, 14.31% of the clips produced by silence-based segmentation begin mid-sentence and 12.81% do so with no signal that would let a consumer detect it; the sentence-aligned arm leaves no unflagged broken clip at all. Over the full corpus, 23 of the 1,547,494 clips in the recommended subset begin mid-sentence, a rate of 15 per million.

Second, no threshold removes a clip. Every measurement stage writes columns only, and the accept/exclude decision is made by a separate, versioned policy read from configuration, so a consumer can reproduce a stricter or looser subset without re-running a model. This separation is what let us test thresholds instead of assuming them, and two standard ones did not survive: in blind listening, a clipping-ratio gate rejected clips that listeners judged usable 25 times out of 25, and an ASR-confidence gate rejected 15 of 30 clips whose transcripts were verbatim correct. Both rules were removed from the policy and both columns are still published.

We report the corpus, the measured limits of its source material, a 90-speaker clustering derived without fixing the number of speakers, an evaluation split with zero recording and text leakage, and the pipeline cost on a single GPU.

1 Introduction

Text-to-speech systems are trained on read speech: one speaker, a prepared text, a quiet room or a good microphone, and sentences delivered whole. The supply of such material is uneven across languages in a way that does not track the number of speakers. Turkish is a case in point. It has large text corpora, it is covered by the multilingual speech recognition and representation models, and transcribed speech for recognition exists in quantity; what it does not have is read speech in the form TTS training needs. The openly available Turkish read-speech corpora are measured in tens or low hundreds of hours, which is one to two orders of magnitude below what comparable English resources offer.

The gap is therefore not in the language but in the task. This matters for how one should try to close it. A method that only works where a language is already well served is of little use; a method that turns publicly available audiobook channels into a TTS-grade corpus, and that states its own defects in measurable terms, transfers to any language that has such channels. That is what we set out to build, and what we release here alongside the corpus itself.

Contributions.

1. **The corpus.** 1,840,404 clips, 3,105.7 hours of Turkish read speech at 24 kHz from 2,680 recordings on 27 channels, with 90 clustered speaker identities, published with every per-clip measurement the pipeline computed and with an evaluation split whose recording and text leakage are both zero (Section 3).
2. **Segmentation at sentence boundaries, measured against the alternative.** We reverse the usual order — long-form ASR with word timestamps first, sentence segmentation on the transcript second, cutting third — and quantify what that buys in a controlled ablation where the cut point is the only variable (Section 7.1).
3. **A gate-free, versioned release policy.** Measurement stages are structurally forbidden to make decisions; the recommended subset is produced by a policy that lives in configuration, carries a version number, and records a reason for every exclusion (Section 6).
4. **Validated negative results.** Two thresholds in common use failed blind listening on this corpus and were removed. We report the protocol, the listener verdicts, and the columns that remain published so that others can re-impose the thresholds if their own listening supports them (Section 7.3).
5. **A method for multi-hour recordings.** 30.7% of our source material is in single recordings longer than four hours, and the longest is 14.9 hours. Processing these under a fixed memory budget shapes the whole architecture (Sections 4 and 8).

2 Related work

The established pipeline. Contemporary pipelines that turn found audio into speech corpora share a skeleton: standardisation, optional source separation, speaker diarisation, segmentation by voice activity detection, ASR, and quality filtering. Emilia [1, 2] is the clearest statement of it, and GigaSpeech 2 [10] instantiates the same order for languages with less data, transcribing with Whisper [8] and force-aligning with MMS [7]. We share most of this skeleton and depart from it in two places.

Where to cut. Cutting at sentence boundaries rather than at silences is not new, and we do not claim it as a contribution: LibriTTS [11] separated itself from LibriSpeech [6] on exactly this point, Libriheavy [3] places its cuts at sentence boundaries as well, and WenetSpeech4TTS [5] attacks the same defect from the other side by merging adjacent segments and widening boundaries so that words are not cut. What is missing from the literature, and what we supply, is a controlled measurement of the cost of the alternative on the same recordings, including the part that matters most to a consumer: whether the resulting defect is detectable in the released metadata.

Filtering as a gate. Pipelines that build data to feed their own model filter inside the pipeline with fixed thresholds. Emilia discards clips below a DNSMOS [9] OVRL of 3.0; LibriTTS requires an estimated SNR above 20 dB for its “clean” subset, and its stricter path reduces LibriSpeech’s 982 hours to 585. Projects that release a shared corpus more often score and stratify instead: WenetSpeech4TTS publishes Premium, Standard and Basic tiers cut at DNSMOS P.808 thresholds and ships all three, and LibriTTS-R [4] takes the same instinct further by restoring clips rather than discarding them. We go one step beyond stratification: nothing is removed at any tier, the thresholds live outside the data, and the policy that applies them is versioned so that a published number can be reproduced exactly or deliberately changed.

recording length	recordings	hours	share of corpus
< 10 min	168	20.4	0.6%
10–30 min	911	274.0	8.0%
30–60 min	635	462.8	13.5%
1–2 h	467	654.7	19.0%
2–4 h	351	973.4	28.3%
4–8 h	131	700.5	20.4%
8 h and over	35	354.4	10.3%

Table 1: Length distribution of the 2,698 readable source recordings.

Turkish read speech. The Turkish speech resources in open circulation are dominated by recognition-oriented material: read prompts collected by volunteers, broadcast news, and multilingual evaluation sets. The read-speech portion suitable for TTS training is small in every case, which is the gap this release addresses.

3 Source material and its measured limits

The raw material is 2,700 files downloaded from 27 public Turkish audiobook and spoken-article channels. Every file was probed before any processing began: 2,698 are readable audio (one m4a is corrupt, one file is a README), totalling **3,440.2 hours** and 200.1 GB.

Recordings are long. The median recording is 41.4 minutes and the longest is 14.92 hours. Table 1 gives the distribution. The consequential figure is that **30.7% of the corpus hours sit in single recordings longer than four hours**. Processing a multi-hour recording under a fixed memory budget is therefore a requirement of this material, not an optimisation, and it shapes the pipeline described in Section 4. The established pipelines do not help here: Emilia’s inputs range from 20 seconds to just under one hour, and Libriheavy resolves long-form audio by aligning it to the book text, a reference we do not have.

The encoding profile is uniform, and that is a limitation. 2,618 files are AAC (3,379.7 hours) and 80 are MP3 (60.5 hours); 2,693 are at 44.1 kHz; the bit rate is tightly concentrated (5th percentile 128, median 129, 95th percentile 130 kb/s). We measured the true bandwidth of the source by taking a 60-second excerpt from the middle of one recording per channel and computing a 32k-point FFT: the median cutoff frequency is **15.7 kHz** (minimum 13.0, maximum 15.8), and the median share of energy above 12 kHz is 0.05%. A 24 kHz release rate (12 kHz Nyquist) therefore sits below the ceiling of the source everywhere in the corpus; upsampling further would buy nothing and downsampling has no justification. The uniformity itself is a limitation of the corpus and we state it in Section 9: a model trained on KIRAAT sees one codec profile.

Channel shares are unequal, and no cap was applied. The largest channel contributes 19.5% of the raw hours and the second 15.5%; the top two together contribute 35.2%. We considered a per-channel hour cap and decided against it: the cap that would have equalised the channels would have reduced the corpus from 3,440 to 1,849 hours. Pruning channel diversity with a threshold is precisely the move this pipeline exists to avoid, and the imbalance is published rather than hidden — every clip carries its `channel` and `duration`, so a consumer can impose any cap they like.

stage	what it does	model / setting
prepare	decode, 24 kHz mono, 40 Hz high-pass, -1 dB peak	ffmpeg
asr	long-form transcription with word timestamps	faster-whisper <code>large-v3</code> , rev. <code>edaa852e</code> , beam 5, temperature 0, <code>condition_on_previous_text: false</code>
align	forced alignment, per-word score	torchaudio <code>MMS_FA</code> (with <code><star></code>)
boilerplate	channel-level mining of station idents and announcements	word n -grams, threshold 40% of a channel's recordings (min. 2), <code>max_words: 24</code>
segment	cut at sentence boundaries, audio-based boundary refinement	<code>kiraat/segment.py</code> , <code>kiraat/boundaries.py</code>
clip_qc	speech ratio, internal silence, edge silences, level, LUFs	silero VAD; BS.1770
music	music-to-speech level and AudioSet music score	torchaudio <code>HDEMUCS_HIGH_MUSDB_PLUS</code> ; MIT/ <code>ast-finetuned-audio</code> set-10-10-0.4593
dnsmos	P.835 SIG / BAK / OVRL	DNS-Challenge <code>sig_bak_ovr.onnx</code> , sha256 pinned in config
speaker	per-recording voice identity and within-recording consistency	ECAPA-TDNN <code>speechbrain/spkrec-ecapa-voxceleb</code> , rev. 5c0be387
export	manifest and run record	<code>manifests/clips.jsonl</code> , <code>run.json</code>

Table 2: Pipeline stages. Every model revision is pinned and recorded in `run.json` together with the git commit, the stage versions and the configuration hash.

4 Pipeline

The pipeline runs as a sequence of stages over a SQLite state store, with CPU and GPU work overlapped by a scheduler that keeps one writer. Table 2 lists the stages and the models, with the weight revisions pinned in the run record.

4.1 Segmentation at sentence boundaries

The order is the point. A pipeline that segments first and transcribes second can only cut where the audio is quiet, and quietness does not mark sentence ends: a narrator breathes at a comma and may not breathe between two sentences. We transcribe the whole recording first, obtain word-level timestamps, segment the *transcript* into sentences, and only then cut the audio at the timestamps of the sentence boundaries. Voice activity detection is retained, but only to find the speech regions handed to ASR.

Two situations force a cut that is not a sentence boundary. A sentence longer than the duration ceiling is split at its internal punctuation, and the resulting clips carry the `forced_split` flag; a gap inside a sentence that exceeds the tolerated silence forces a split flagged `gap_split`. The flags are the mechanism by which the defect stays visible: Section 7.2 shows that 97.6% of all mid-sentence starts in the corpus carry one of these two flags, and that the policy therefore keeps them out of the recommended subset.

4.2 Boundary refinement

Cutting at a sentence boundary fixes *which* point to cut at, not *exactly where* that point is: a forced aligner reports the end of a word slightly early, so a cut placed on the timestamp clips the final phoneme. The refinement stage looks at signal energy in a small window around the

definition	clips counted as duplicates
text + channel (the obvious key)	72,839
every measured value identical	0
audio measurements identical, within channel	592
audio measurements identical, channel ignored	598
duration rounded to 0.1 s, LUFS to 0.1 dB	15,752

Table 3: Duplicate counts over all 1,840,404 clips under five definitions. The second row shows that a literal reading of the rule is unusable: a channel’s per-episode ident is byte-identical audio, but `word_confidence` and `leading_silence_sec` differ because they measure the clip’s surroundings, not its audio.

timestamp between adjacent clips and pulls the cut to the actual silence, with the window and thresholds in configuration.

This was found by ear, not by a metric. A first listening round found truncated syllables; the refinement was written in response; a third round on 28 clips found 28 clean — zero broken starts, zero broken ends, zero truncated syllables — and a fourth round on 30 clips (20 suspected, 10 controls, shuffled) found zero truncated words in all of them. The audit criterion moved with the finding: it is no longer “does the cut match the timestamp” but “does any word inside the clip’s own word span fall outside the clip”, that is, does the text match the audio.

4.3 Duplicates

A duplicate is not the same text read twice. Two narrators reading the same sentence is prosodic diversity and is kept; a duplicate is the same text *and* the same audio. We define “same audio” by measurement — duration, LUFS, RMS and peak matching exactly — and deliberately do not put the channel in the key, since matching measurements mean the same recording regardless of where it was published. Table 3 shows why the definition had to be measured rather than assumed: the obvious key over-counts by two orders of magnitude, and a tolerant key collapses genuinely different readings.

The released manifest flags **576** clips as duplicates, 453 of them in a single channel’s episode ident. 150 clips have no audio identity because their audio could not be read, and are passed through without a duplicate check.

5 Measurements, not decisions

Every clip stage writes columns and nothing else. This is enforced rather than documented: the stage base class rejects any stage output containing a `recommended` or `decision` field. Thresholds do not appear in code at all; they live in a single configuration file whose loader refuses to silently accept an unknown section or key. The column schema has one definition in the source, and the dataset card is generated from it, so the published documentation cannot drift from the published columns.

The consequence for a consumer is concrete. Every clip in the release carries its speech ratio, internal and edge silences, peak, RMS and BS.1770 loudness, clipping ratio, AudioSet music score and separated music-to-speech level, DNSMOS SIG/BAK/OVRL, mean and minimum forced-alignment score, ASR word confidence, word count, speaker identity, within-recording speaker consistency, and its flags — whether or not our policy recommends it. Re-deriving a different subset requires re-reading a manifest, not re-running a GPU stage.

Audio is measured, not modified. Source separation is run to *measure* background music, and its output is discarded; the released waveform is the decoded source at 24 kHz, and loudness

measure	sentence-aligned	silence-aligned
clips	141,379	76,466
hours	240.54	254.18
begins mid-sentence	1,967 (1.39%)	10,943 (14.31%)
ends mid-sentence	2,405 (1.70%)	12,117 (15.85%)
unflagged broken start	0 (0.00%)	9,798 (12.81%)
unflagged broken end	0.01%	11.59%
broken at both ends	418	2,673
median duration	5.94 s	12.90 s
duration, 5th–95th pct	2.69–10.96 s	5.10–15.40 s
at the duration ceiling	0.64%	14.70%
mean <code>align_score</code>	0.9442	0.9459
mean of per-clip <i>minimum</i> <code>align_score</code>	0.7319	0.6469

Table 4: Segmentation ablation over 206 recordings (240.5 hours) from all 27 channels. Only the cut point differs between the two arms.

is a published column rather than an applied normalisation. Pipelines that normalise level and publish the separated signal make a choice on the consumer’s behalf that cannot be undone downstream; we make it undoable by not making it.

6 The recommended subset policy

The policy is a versioned object in configuration. Version 6, the one used for this release, has three rules: `speech_ratio` ≥ 0.60 , `internal_silence_sec` ≤ 1.0 , and the absence of any of the flags `oversize`, `forced_split`, `gap_split`, `short`, `duplicate`, `boilerplate`, `background_music`. Applying it to the full corpus yields the recommended subset: **1,547,494 clips, 2,575.2 hours**, which is 84.1% of the clips and 82.9% of the hours. Every excluded clip carries the reason it was excluded.

Because the policy is separate from the data, its history is legible, and its history is mostly a history of rules that were *removed*. That is the subject of the next section.

7 Results

7.1 Segmentation: what the cut point costs

We ran both segmentation strategies over the same recordings, from the same word timestamps, with the same duration settings; the only variable is where the cut is placed. The sentence arm is the output of the full pipeline; the baseline arm cuts the speech regions found by silero VAD at the silences between them, splitting hard at the ceiling when a single region exceeds it. The sample is channel-balanced: 206 recordings from all 27 channels, 240.5 hours. Table 4 gives the result.

The row that matters is the unflagged one. In the sentence arm there is *no* unflagged clip with a broken start: all of the 1.39% carry `forced_split` or a boilerplate flag, so the defect is visible in the release and excluded from the recommended subset. In the baseline arm almost all of the breakage is unflagged (12.81%), because a pipeline that cuts at silence has no signal with which to mark it. The defect varies by channel but never disappears: the baseline arm’s mid-sentence rate ranges from 7.0% to 35.0% across channels, the sentence arm’s from 0.00% to 5.02%.

Two side findings. Silence-aligned segmentation yields *more* hours (254.2 against 240.5) from

subset	clips	hours	broken start	broken end
whole corpus	1,840,404	3,105.7	27,483 (1.49%)	33,346 (1.81%)
recommended subset	1,547,494	2,575.2	23 (0.0015%)	430 (0.03%)
outside the policy	292,910	530.5	27,460 (9.37%)	32,916 (11.24%)

Table 5: Sentence integrity over the full corpus. Rates are per clip.

roughly half as many clips: its clips are twice as long and 14.7% of them end at the duration ceiling, meaning the cut point is set by the ceiling rather than by the content. And while mean alignment confidence is identical across the two arms, the mean of the per-clip *minimum* word confidence is clearly higher in the sentence arm (0.732 against 0.647): clips cut at silences carry more badly aligned words at their edges.

7.2 Sentence integrity over the full corpus

The ablation is a controlled comparison on a balanced sample. We also applied the same two definitions — a clip whose text begins with a lowercase letter, and a clip whose text does not end in sentence-final punctuation, both computed with the same Turkish-aware routines — to all 1,840,404 clips.

Table 5 should be read together with the flags. Of the 27,483 clips that begin mid-sentence, 97.6% carry `forced_split` or `gap_split` (92.3% carry `forced_split` alone); among the 1,558,409 clips that carry no flag at all, 23 begin mid-sentence, a rate of 15 per million. The corpus does contain mid-sentence cuts — a sentence that runs past the duration ceiling has to be split somewhere — but they are cuts the pipeline declares, and declaring them is what allows the policy to exclude them and a consumer to find them.

7.3 Blind listening: two standard thresholds that did not hold

Because the thresholds are separate from the data, they can be tested. Our protocol hides the channel and the measurement, balances the clips across bands of the measurement under test, and records the listener’s verdict to a file before the scores are revealed. Six rounds were run: two on background music, two on boundary cuts, one on clipping ratio and one on ASR word confidence. Two of them ended with a rule being deleted.

Clipping ratio. A rule excluding clips with a clipping ratio above 0.002 was removing 536 clips — and 532 of them were in a single channel, 93.2% of that channel’s output. That is the signature of a channel property, not a clip defect. A blind round of 25 clips balanced across six bands of the measure (0 to 0.0227) found 24 “no audible distortion” and one “slight”, and that one was in the lowest band; all 25 were judged fit for training. The rule was removed in policy v5.

ASR word confidence. A rule requiring word confidence ≥ 0.60 was tested the same way: 30 clips across six bands from 0.18 to 0.99, channel and score hidden. In 29 the transcript was verbatim correct; the single error was in the lowest-confidence clip. The threshold would have removed 15 of these 30, of which 14 were found flawless. An independent check without ears agrees that the signal is weak but real: rank correlation with the aligner score is +0.262, and even in the worst band the median `align_score_mean` is 0.938. The rule was removed in policy v6, and the column is still published as a ranking signal.

What survived. The background-music rule survived the same treatment and was corrected by it. Our first guess of -30 dB was refuted: in the -40 to -33 dB band, 6 of 6 clips had clearly

column	p05	p50	p95	mean
duration (s)	2.76	5.82	10.56	5.99
loudness_lufs	-27.99	-20.92	-12.54	-20.61
dnsmos_ovrl	2.862	3.296	3.491	3.253
dnsmos_sig	3.271	3.563	3.710	3.536
dnsmos_bak	3.631	4.109	4.210	4.041
speech_ratio	0.851	0.977	1.000	0.955
music_score_audioset	0.001	0.006	0.228	0.034
music_to_speech_db	-80.0	-80.0	-40.94	-75.42
align_score_mean	0.858	0.961	0.994	0.946
align_score_min	0.278	0.800	0.976	0.740
word_confidence	0.550	0.917	0.999	0.866
n_words	5	11	21	11.7
internal_silence_sec	0.000	0.000	0.664	0.166

Table 6: Column distributions over the recommended subset (1,547,494 clips).

audible music. A second round of 36 clips balanced across four level bands found audible music in 86% of the flagged clips and clean audio in 11 of 14 unflagged ones. The threshold stayed at -40 dB, and the rule is the conjunction of two columns — separated music-to-speech level above -40 dB *and* AudioSet music score ≥ 0.3 — because the two columns measure different things: their Spearman correlation on 3,097 separated clips is only $+0.479$.

A rule we did not re-impose. The DNSMOS OVRL ≥ 3.0 gate used by other pipelines is not in our policy. The column exists for every clip and 83.57% of the corpus would pass it; we report that number precisely so that the gate’s effect is visible, and we will not adopt it until it passes the same blind listening the other rules were subjected to. The burden of proof is on the gate, not on the data.

7.4 The released columns

Table 6 gives the distribution of the main columns over the recommended subset. Two of them carry design decisions. Loudness is published and *not* applied: the median clip is at -20.92 LUFS with a 5th–95th percentile range of -27.99 to -12.54 , and normalising is the consumer’s choice. DNSMOS is published for every clip: over the whole corpus the mean OVRL is 3.208 (5th percentile 2.69, median 3.27, 95th 3.49).

The flags over the whole corpus are: `background_music` 207,196, `forced_split` 52,501, `short` 20,904, `oversize` 6,731, `gap_split` 5,141, `boilerplate` 1,053, `duplicate` 576. The music flag alone accounts for most of the difference between the corpus and the recommended subset.

7.5 Speakers

Channels are not speakers, and we did not assume they were. The `speaker` stage computes an ECAPA-TDNN centroid per recording and a within-recording consistency; the identity decision is made afterwards, at corpus level, by average-linkage clustering. The merge threshold is not chosen by hand and the number of clusters is never supplied: we measure the similarity distribution within recordings and across recordings, and take the threshold at the equal error point of the two.

On the full corpus (2,677 recordings; 21 have no clip long enough to establish an identity) the equal error point is at **0.532** with an error of 0.52%, and clustering at that threshold yields **90 speakers**. The median margin is 0.445 and no recording has a negative margin.

split	clips	hours	recordings	channels
train	1,512,448	2,517.08	2,579	27
dev	5,750	9.54	40	27
test	5,473	9.30	45	27
rest	316,733	569.78	—	27
total	1,840,404	3,105.70	2,680	27

Table 7: The four published splits. **train**, **dev** and **test** are drawn from the recommended subset; **rest** carries every remaining clip, so the four splits together are the whole corpus. The recordings column counts recordings whose clips form a split; **rest** draws clips from recordings across all three, and 16 of the 2,680 recordings have all of their clips in it.

Two findings bear on the design. **13 of the 27 channels contain more than one speaker**, and **9 clusters span several channels** — the same narrator publishes on two or three of them. Channel balance is therefore not speaker balance, and neither is a substitute for the other. The distribution is long-tailed: the largest speaker holds 18.2% of the hours, the top five 54.4% and the top ten 75.0%, while 17 speakers have less than an hour.

7.6 Evaluation split

The split is at the recording level: all clips of a recording go to one split, so evaluation is on a recording the model has never heard. The **dev** and **test** splits are channel-balanced, because the corpus is not and an evaluation set that inherited the imbalance would report the numbers of two channels. Recordings run for hours, so dev and test are reduced to a per-channel hour target and the clips beyond that target are left out of training rather than moved into it; they are published in the **rest** split described below.

Nothing is left out of the release. The clips that the policy does not recommend, and the recommended clips that dev and test did not use, are published in a fourth split, **rest**: 292,910 clips (530.5 hours) with **recommended=false**, and 23,823 clips (39.3 hours) that are recommended but fell beyond a channel’s dev/test duration target or were dropped by the text-leakage cleanup below. A clip in **rest** carries exactly the same measurement columns as one in **train**, so a consumer who sets a different policy can recover it from the release rather than re-running the pipeline — which is the whole point of not gating.

Recording leakage is zero by construction and is nevertheless tested. Text leakage was not zero by construction: 2,263 test clips and 2,723 dev clips carried a sentence that appears verbatim in training, because the same book is read by different narrators on different channels. They were dropped from the evaluation sets — training was left untouched — and the remaining leakage is zero.

Speaker overlap is a different matter and we report it plainly: train has 89 speakers, dev 26 and test 26, and **25 of the 26 test speakers also occur in training**. This is the usual setup for TTS evaluation, but it is a *seen-speaker* evaluation and must not be described otherwise. An unseen-speaker evaluation would require a separately constructed subset.

8 Cost and reproducibility

The full corpus was processed on one machine: an RTX 3090 (24 GB) with 12 CPU workers, under a 280 W power cap. Forced alignment is the bottleneck at 94 audio-hours per hour with a single worker. Profiling it showed 56% of the time in the model forward pass, 28% in token merging and 10% in the alignment itself; the token-merging cost was a GPU synchronisation per token span, and moving the score tensor to CPU raised the same recording from 109× to

188× real time with byte-identical output. The other stages measured 46 clips/s for clip QC across 12 CPU workers, 28 clips/s for music, 81.1 clips/s for DNSMOS and about one second per recording for the speaker stage.

The working directory occupies 567 GB: 284 GB of intermediate 24 kHz mono FLAC, 246 GB of clips (about 159 KB per clip), 2.7 GB of alignment, 1.5 GB of ASR output and a 1.4 GB state database.

Every run writes a record containing the git commit, the version of each stage, the configuration hash and the pinned revision of every model weight, so that a published number can be traced to the code and the weights that produced it.

9 Limitations

No downstream TTS experiment yet. This paper describes and measures a corpus and the pipeline that built it. It does not yet report a TTS system trained on that corpus. The comparison that would close the argument — equal-budget models trained on the recommended subset and on the whole corpus, evaluated with MOS/CMOS, intelligibility and speaker similarity — is designed but not run, and until it is, the claims here are claims about data, not about synthesis quality.

Numerals are not force-aligned. The aligner’s lexicon has no digits, so words containing numerals fall back to the ASR timestamp. Rewriting them into their spoken form before alignment was rejected because it would alter the published text.

Alignment noise at sentence starts. 245 words in 289 clips (2.12% of a probed sample) receive implausible alignments on short sentence-initial words. The cause is not established; a chunk-boundary hypothesis was tested and rejected.

Channel and speaker concentration. The top two channels are 38.2% of the recommended subset and the top five speakers are 54.4% of the hours. Both are published per clip so that a consumer can rebalance, but the corpus as released is concentrated.

One encoding profile. The entire corpus is roughly 129 kb/s AAC with a median source bandwidth of 15.7 kHz, and two channels are band-limited near 13 kHz.

Seen-speaker evaluation. 25 of the 26 test speakers appear in training, as reported above.

Boilerplate detection is n-gram based. Verbatim *n*-gram mining does not catch templated announcements with variable slots (“from the book *X* by *Y*”). 23 clips in the recommended subset still begin mid-sentence, and they come from this path.

10 Conclusion

KIRAAT is 3,105.7 hours of Turkish read speech, of which 2,575.2 hours are recommended by a policy that anyone can re-run, tighten or discard. The two rules that shaped it are simple to state and were expensive to establish: cut where sentences end, and let no threshold delete a clip. The first is measurable — on the same recordings, silence-based segmentation leaves 12.81% of clips broken and undetectable, against none in the sentence-aligned arm. The second is what allowed the thresholds themselves to be tested, and two standard ones did not survive contact with a listener.

Availability

References

- [1] Haorui He, Zengqiang Shang, Chaoren Wang, Xuyuan Li, Yicheng Gu, Hua Hua, Liwei Liu, Chen Yang, Jiaqi Li, Peiyang Shi, Yuancheng Wang, Kai Chen, Pengyuan Zhang, and Zhizheng Wu. Emilia: An extensive, multilingual, and diverse speech dataset for large-scale speech generation. In *Proc. IEEE Spoken Language Technology Workshop (SLT)*, 2024. arXiv:2407.05361.
- [2] Haorui He, Zengqiang Shang, Chaoren Wang, Xuyuan Li, Yicheng Gu, Hua Hua, Liwei Liu, Chen Yang, Jiaqi Li, Peiyang Shi, Yuancheng Wang, Kai Chen, Pengyuan Zhang, and Zhizheng Wu. Emilia: A large-scale, extensive, multilingual, and diverse dataset for speech generation. *arXiv preprint arXiv:2501.15907*, 2025.
- [3] Wei Kang, Xiaoyu Yang, Zengwei Yao, Fangjun Kuang, Yifan Yang, Liyong Guo, Long Lin, and Daniel Povey. Libriheavy: A 50,000 hours ASR corpus with punctuation casing and context. In *Proc. IEEE ICASSP*, 2024. arXiv:2309.08105.
- [4] Yuma Koizumi, Heiga Zen, Shigeki Karita, Yifan Ding, Kohei Yatabe, Nobuyuki Morioka, Michiel Bacchiani, Yu Zhang, Wei Han, and Ankur Bapna. LibriTTS-R: A restored multi-speaker text-to-speech corpus. In *Proc. Interspeech*, 2023. arXiv:2305.18802.
- [5] Linhan Ma, Dake Guo, Kun Song, Yuepeng Jiang, Shuai Wang, Liumeng Xue, Weiming Xu, Huan Zhao, Binbin Zhang, and Lei Xie. WenetSpeech4TTS: A 12,800-hour mandarin TTS corpus for large speech generation model benchmark. In *Proc. Interspeech*, 2024. arXiv:2406.05763.
- [6] Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. Librispeech: An ASR corpus based on public domain audio books. In *Proc. IEEE ICASSP*, pages 5206–5210, 2015.
- [7] Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayani Kundu, Ali Elkahky, Zhaocheng Ni, Apoorv Vyas, Maryam Fazel-Zarandi, Alexei Baevski, Yossi Adi, Xiaohui Zhang, Wei-Ning Hsu, Alexis Conneau, and Michael Auli. Scaling speech technology to 1,000+ languages. *Journal of Machine Learning Research*, 25, 2024. arXiv:2305.13516.
- [8] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *Proc. International Conference on Machine Learning (ICML)*, 2023. arXiv:2212.04356.
- [9] Chandan K. A. Reddy, Vishak Gopal, and Ross Cutler. DNSMOS P.835: A non-intrusive perceptual objective speech quality metric to evaluate noise suppressors. In *Proc. IEEE ICASSP*, 2022. arXiv:2110.01763.
- [10] Yifan Yang, Zheshu Song, Jianheng Zhuo, Mingyu Cui, Jinpeng Li, Bo Yang, Yexing Du, Ziyang Ma, Xunying Liu, Ziyuan Wang, Ke Li, Shuai Fan, Kai Yu, Wei-Qiang Zhang, Guoguo Chen, and Daniel Povey. GigaSpeech 2: An evolving, large-scale and multi-domain ASR corpus for low-resource languages with automated crawling, transcription and refinement. *arXiv preprint arXiv:2406.11546*, 2024.
- [11] Heiga Zen, Viet Dang, Rob Clark, Yu Zhang, Ron J. Weiss, Ye Jia, Zhifeng Chen, and Yonghui Wu. LibriTTS: A corpus derived from LibriSpeech for text-to-speech. In *Proc. Interspeech*, pages 1526–1530, 2019. arXiv:1904.02882.